

Monash University

FIT5202 – Data processing for Big Data 2026 S1

Assignment 2B: Using Streaming and ML for Traffic Accident Prediction and Visualisation

Due: **(23:55 Monday 08/Jun/2026, Week 14)**

Weight: 15%

Background (The same as A2A)

Road safety is a paramount concern for modern smart cities. By understanding the underlying patterns that contribute to traffic accidents, local authorities can optimise emergency response times, improve road infrastructure, and implement dynamic traffic calming measures.

In a traffic accident, the "Golden Hour" refers to the critical 60-minute window following a traumatic injury, during which prompt medical treatment offers the best chance of survival. In a modern metropolitan transport network, road safety is a paramount concern. However, when an accident first occurs, the initial reports—often triggered by automated vehicle telemetry, roadside cameras, or panicked civilian emergency calls—are chaotic and lack precise details on severity.

Emergency dispatchers, with limited resources, face a critical resource allocation problem: should they send a standard patrol car to assess the situation or dispatch multiple ambulances, fire crews, and heavy rescue vehicles immediately? Sending too much leaves the city vulnerable elsewhere; sending too little costs lives.

Big data processing enables us to analyse vast amounts of data and identify the key drivers of serious accidents. We can use a detailed historical record of road accidents, including environmental conditions, road characteristics, and vehicle manoeuvres, to build models that assist emergency responses.

Objective of the Project

In this assignment, you will act as a data scientist for a national transport authority. In the previous assignment (A1), you used Apache Spark to study historical data; it might have given you some insight into the data set, for example, which feature/attribute contributes to a fatal accident. Then, we built a model to predict the severity of an accident based on environmental, human, and road conditions (assignment 2A). In this assignment (2B), you will use your trained model to build a **real-time** smart-city intelligence dispatch **dashboard** for the emergency response centre.

Key Information

Submission links can be found in Moodle.

A2B has a **compulsory** interview/demo component to be conducted in weeks 14-15. **The teaching team only marks A2B submissions during your demo session. Failure to attend this demo will result in 0 marks (for A2B).**

(We will announce the demo sign-up sheet in Week 12 or 13. Please pay attention to the unit announcement in the final teaching week. If you have an extension/special consideration, more demo sessions will be arranged after the exam. However, this may delay your final result release.)

The Datasets:

- streaming_collision.csv
- vehicle.csv
- Metadata is the same as A2A (some useless columns may have been removed to reduce dataset size).

What you need to achieve

A2B Use Case	Use streaming data to perform real-time prediction and visualise the results for the accident response centre.	Spark Structured Streaming
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Architecture

The following figure represents the overall architecture of the assignment setup.

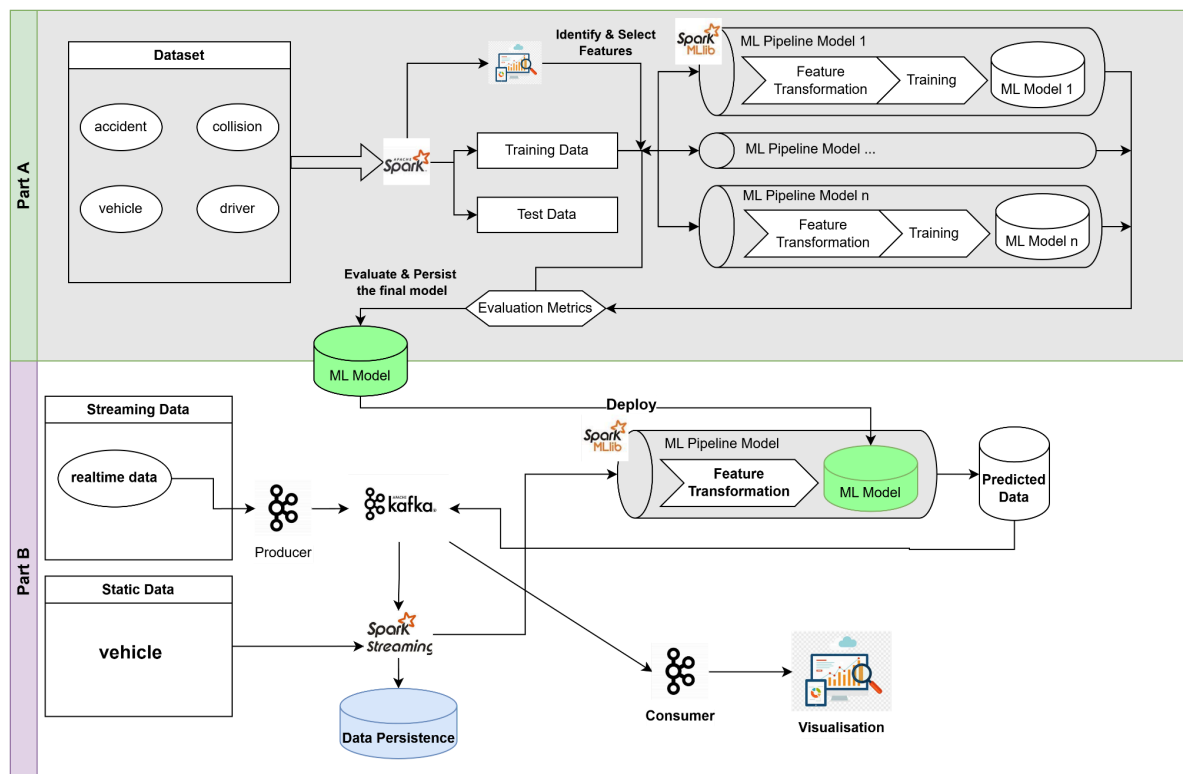


Fig. 1: Overall Architecture for Assignment 2

In both parts, you must implement the solutions using PySpark DataFrame/MLlib for the data preprocessing and machine learning pipelines. Please follow the steps to document the processes and write the code in your Jupyter notebook.

In **Part B**, you will use the model from Part A to generate operational insights for the grid operator.

Getting Started

- Download the datasets from Moodle.
- Download a template file for submission purposes:
 - **A2B-Task1_producer.ipynb** file for streaming data production
 - **A2B-Task2_spark_streaming.ipynb** file for consuming and processing data using Spark Structured Streaming
 - **A2B-Task3_consumer.ipynb** file for consuming the data using Kafka and visualising
- For submission, please append your authcate ID at the end of the filename. (e.g. **A2B-Task1_producer_xxxx0000.ipynb** **xxxx0000** is your authcate ID.
- You will use Python 3+ and PySpark 3.5.0+ for this assignment (This environment is the same as we used in labs.)

IMPORTANT:

Please answer each question in your Jupyter Notebook file using code/markdown cells. Acknowledge any ideas or codes you referenced from others in the markdown cell or reference list.

If you use generative AI tools, all prompts you use should also be included in the reference section or appendix; alternatively, you can also submit an additional PDF file.

A2 Part B Specification

Your task in this application is to build a visualisation dashboard for the emergency dispatch operator to oversee incoming road traffic accidents and their predicted severity so that they can optimise their emergency resource allocation.

Part 1: Producing the data (10%)

In this task, we will implement Apache Kafka producers to **simulate** real-time data streaming. Spark and parallel data processing should not be used in this section, as we are simulating sensors that often lack processing capabilities.

1. Every 1 second, load 50-100 accident records from the CSV file. You should keep a pointer in the file reading process and advance it per read. **The data should be read**

in chronological order; just the number of records you read every second is a random number between 50 and 100 (inclusive).

2. Add the current timestamp (**accident_ts**) to the **records**.
3. Send your batch of accident data to a Kafka topic with an appropriate name.

Save your code in **Assignment-2B-Task1_producer.ipynb**.

Part 2. Streaming application using Spark Structured Streaming (40%)

In this task, you will implement Spark Structured Streaming to consume the data from task 1 and perform a prediction.

Important:

- This task uses PySpark Structured Streaming with PySpark Dataframe APIs and PySpark ML.
- You also need your pipeline model from A2A to make predictions and persist the results.
- Note for the prediction related to event time: in a real scenario, you should use **accident_ts** as the event time; however, since we are simulating streaming and your model was trained with the time column as a feature, you can choose to use the time column or **accident_ts**.

Tasks:

1. Write code to create a SparkSession, which 1) uses **four cores** with a **proper application name**; 2) uses the UK/London timezone; 3) ensures a checkpoint location has been set.
2. Write code to define the data schema for the data files. Load the static datasets into data frames. (You can reuse your code from 2A.) In a car accident, we collect streaming information like the real-time road condition, but the vehicle information is static and can be read from the vehicle registration database.
3. Using the Kafka topic from the producer in Task 1, deserialize and ingest the streaming data into Spark Streaming, assuming all data comes in the **string** format except for the 'accident_ts' column as a **numeric** type. Then the data frames should be converted to the appropriate types.
4. Use a watermark on **accident_ts**. If data points are received 30 seconds late, discard the data. (note: in a local environment like your laptop, late arrival or delayed processing may never happen.)
5. Perform the necessary transformation you used in A2A. (note: every student may have used different features, feel free to reuse the code you have written in A2A. If you built an end-to-end pipeline, you can ignore this task.)
6. Load your pipeline model and perform the following aggregations (using tumbling windows):
 - a) Make predictions and print **high-severity accidents (>7) every 5 seconds**.
 - b) Every 10 seconds, print the total number of accidents for each severity.
 - c) Every 30 seconds, for each **local district with data**, print the total number of low (1-3), medium (4-6) and high severity (7-10) accidents.
7. Save the data from 6 to Parquet files as streams. (Hint: Parquet files support streaming writing/reading. The file should keep updating while new batches arrive.)

8. Read the Parquet files from task 7 as **data streams** and send them to Kafka topics with appropriate names.

(Note: **You shall read the parquet files as a streaming data frame and send messages to the Kafka topic when new data appears in the parquet file.**)

Save your code in **Assignment-2B-Task2_spark_streaming.ipynb**.

Part 3. Consuming data using Kafka and Visualise (30%)

In this task, we will implement an Apache Kafka consumer to consume the data from Part 2. You shall not use (and don't need) Apache Spark in this section.

Important:

- **In this part, Kafka consumers are used to consume the streaming data published from task 2.8.**
- 1. Load static data and streaming data frames from part 2.
- 2. Create a dashboard with 3 plots to visualise the real-time streaming data from part 2:
 - a) A basic plot to show the number of high-severity accidents (line/bar chart).
 - b) A histogram shows the **cumulative** distribution of accidents by severity.
 - c) A bubble map plot similar to [UK Road Traffic Accidents - Crash View](#). (Just the map; no interactive controllers are required. You can use any library (like folium, seaborn, plotly, etc.)

(Note: Ideally, for a dashboard-like user experience, 3 diagrams need to be updated in real-time simultaneously. You may need to do a bit of searching/research to make it work.)

You can refer to the following sites:

<https://samplecode.link/> (a mirror of <https://python-graph-gallery.com/>)

<https://matplotlib.org/>

<https://plotly.com/python/>

Save your code in **Assignment-2B-Task3_consumer.ipynb**.

Part 4: Demo and Interview (20%)

IMPORTANT: The interview is compulsory, and we only mark your A2B during the interview. No marks will be awarded if the interview is not attended. (0 marks for the whole A2B, not just this section).

The demo/interview session details will be announced/arranged in the last teaching week. Please pay attention to the unit announcement email and Ed forum.

Each demo is roughly 10 minutes. You have 5-6 minutes to show your application; then, your marker will ask 3-4 questions to assess your understanding. Please arrive at your allocated demo session a few minutes early and ensure your laptop/application is working correctly.

Note 2: For students who couldn't complete the visualisation part, you can demonstrate the raw console output of your Spark streams, but a maximum score of 15 will apply.

The demo/interview is marked on a 5-level scale:

The demo is working, and the student has a competent understanding	20
Working demo on console and/or partial understanding	15
The demo is not working and/or partial understanding	10
The demo is not working and/or low understanding	5
No attendance or can't answer most of the questions	0

The demo session will be arranged in Week 14 and 15 (Exam week 1 and 2 after the SWOT).

Submission A2B

You should submit your final version of the assignment solution via Moodle. You must submit the following:

- A **zip** file named based on your authcate name (e.g. abcd1234). The zip file should contain
 - ***Assignment-2B-Task1_producer_authcate.ipynb***
 - ***Assignment-2B-Task2_spark_streaming_authcate.ipynb***
 - ***Assignment-2B-Task3_consumer_authcate.ipynb***

The file in submission should be a ZIP file and *not any other kind of compressed folder (e.g. .rar, .7zip, .tar)*. Please do not include the data files in the ZIP file.

Assignment Marking Rubric

Detailed mark allocation is available in each task. For complex tasks and explanation questions, you will receive marks based on the quality of your work.

In your submission, the Jupyter Notebook file should contain the **code and its output**. It should follow *programming standards, readability of the code, and organisation of code*. Please find the PEP 8 -- Style Guide for Python Code for your reference. Here is the link: <https://peps.python.org/pep-0008/> Penalty applies if your code is hard to understand with insufficient comments.

Other Information

Where to get help

You can ask questions about the assignment in the Assignments section in the Ed Forum, which is accessible on the unit's Moodle Forum page. This is the preferred venue for assignment clarification-type questions. *You should check this forum regularly, as the responses of the teaching staff are "official" and can constitute amendments or additions to the assignment specification.* Also, you can attend scheduled consultation sessions if the problem and the confusion are still unresolved.

Searching and learning on commercial websites/forums (e.g. Quora, Stack Overflow) is allowed. However, you should not post/ask assignment questions on those forums.

Plagiarism and collusion

Plagiarism and collusion are serious academic offences at Monash University. Students must not share their work with other students. Students should consult the policy linked below for more information.

<https://www.monash.edu/students/academic/policies/academic-integrity>

See also the video linked on the Moodle page under the Assignment block.

Students involved in collusion or plagiarism will be subject to disciplinary penalties, which can include:

- The work is not being assessed
- A zero grade for the unit
- Suspension from the University
- Exclusion from the University

Late submissions and Special Consideration

ALL Special Consideration, including within the semester, is now handled centrally. This means that students **MUST** submit an online Special Consideration form via Monash Connect. For more details, please refer to the **Unit Information** section in Moodle.

There is a 5% penalty per day, including weekends, for a late submission. Also, the cut-off date is 7 days after the due date. No submission will be accepted (i.e. zero mark) after the cut-off date unless you have a special consideration.

Mark Release and Review

- Mark will be released within 10 business days after the submission deadline.
- Reviews and disputes regarding the mark will be accepted a maximum of 7 days after the release date (including weekends).

Generative AI Statement

As per the University's [policy](#) on the guidelines and practices pertaining to the usage of Generative AI:

AI & Generative AI tools may be used SELECTIVELY within this assessment.

Where used, AI must be used responsibly, clearly documented and appropriately acknowledged (see Learn HQ).

Any work submitted for a mark must:

1. Represent a sincere demonstration of your human efforts, skills, and subject knowledge for which you will be accountable.
2. Adhere to the guidelines for AI use set for the assessment task.
3. Reflect the University's commitment to academic integrity and ethical behaviour.

Inappropriate AI use and/or AI use without acknowledgement will be considered a breach of academic integrity.

The teaching team encourages students to apply their own critical thinking and reasoning skills when working on the assessments with assistance from GenAI. Generative AI tools may produce inaccurate content, which could have a negative impact on students' comprehension of big data topics.